Upper high school · probability and sequences

# Waiting for the first success

Synthesis problem

## Investigate, justify, conclude

Roll a fair die until the first 6, but stop after 10 rolls in any case. Let N be the number of rolls actually made. Find $P(N = 10)$, the probability of obtaining at least one 6, and E(N). For the mean, use $E(N) = \sum P(N > n)$, from $n = 0$ to 9.